

Adaptive Component Screening (ACS) — V4.0

Engineering Intelligence Upgrade: Complete Implementation Plan

Document Purpose

This document defines the complete V4.0 engineering intelligence upgrade while preserving the existing ACS scientific architecture. It is intended as an implementation source of truth for engineering, ML, backend and frontend work, with explicit safety, auditability and information-boundary requirements.

Problem Statement

Smart India Hackathon 2026 — PS 26170

Scope

Centralized metrics, Component Digital Passport, safety margin, decision trace, OOD decomposition, recommendations, lot intelligence, APIs, audit and regression testing

Implementation Basis

Existing ACS anomaly detection, forecasting, OOD, safety-policy, API and frontend foundations

Primary Principle

AI generates evidence; engineering policy and authoritative limits determine disposition

Safety Boundary

AI evidence may inform engineering screening, but it must never override an authoritative engineering specification or silently change the safety policy.

*Section-by-section index of the engineering intelligence upgrade***Contents**

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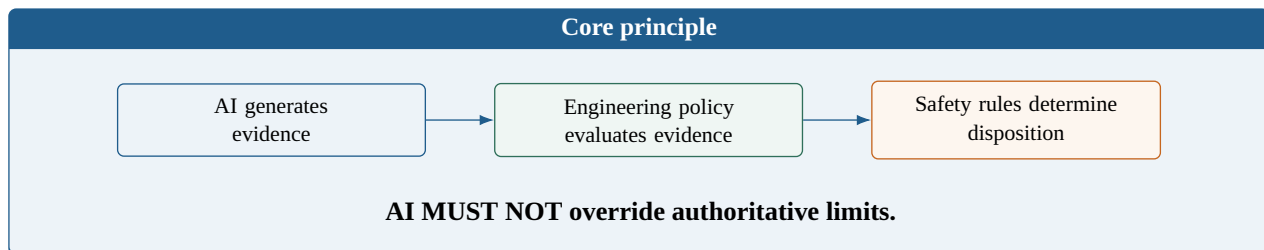
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Upgrade the existing ACS prototype into a complete engineering intelligence platform without breaking the existing working anomaly detection, forecasting, OOD, safety-policy, API, or frontend functionality.

The implementation must preserve the existing ACS scientific architecture while adding:

- Centralized engineering metrics
- Component Digital Passport
- Safety Margin intelligence
- 9-stage Decision Trace
- Enhanced OOD decomposition
- Engineering recommendations
- Lot-level intelligence
- Mission-safety KPIs
- New Screening workflow
- Improved Audit/Benchmark interface
- API extensions
- Automated regression tests
- Stronger frontend/backend integration

The core principle remains:



1 CURRENT SYSTEM FOUNDATION

The existing ACS system already contains the pipeline shown in Figure 1.

The new implementation must build on this architecture rather than replace it.

2 NEW IMPLEMENTATION ARCHITECTURE

The upgraded system becomes the extended pipeline shown in Figure 2.

3 IMPLEMENTATION PRINCIPLES

3.1 Do Not Break Existing Functionality

Before implementing any new feature:

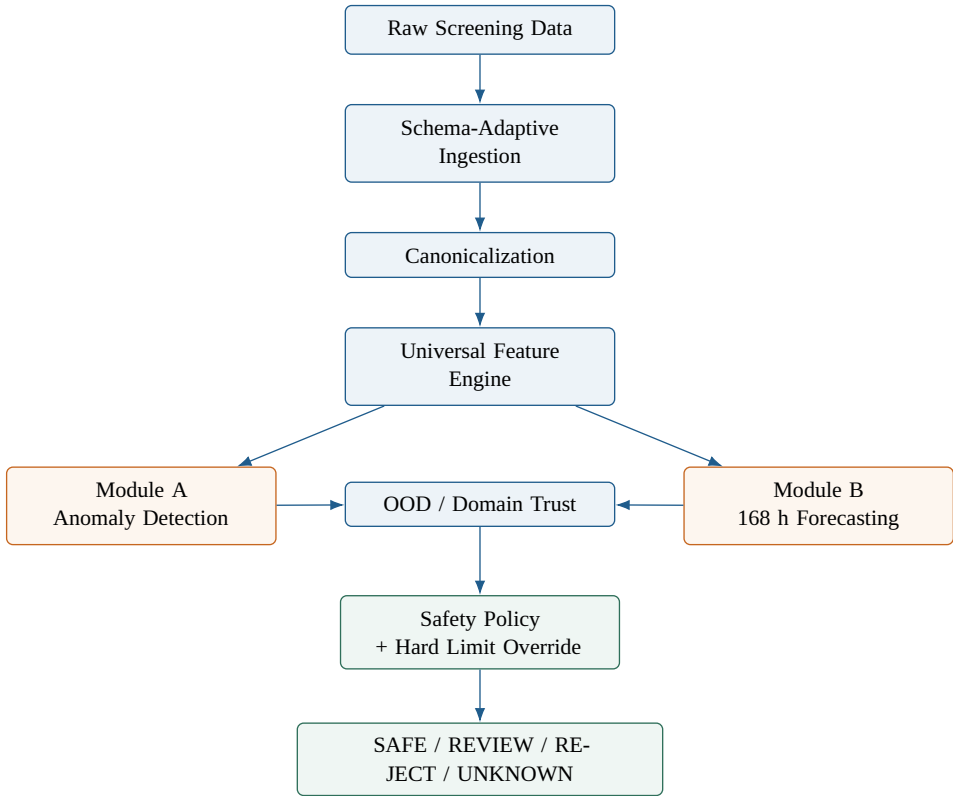


Figure 1: Existing ACS foundation retained as the baseline for the V4.0 upgrade.

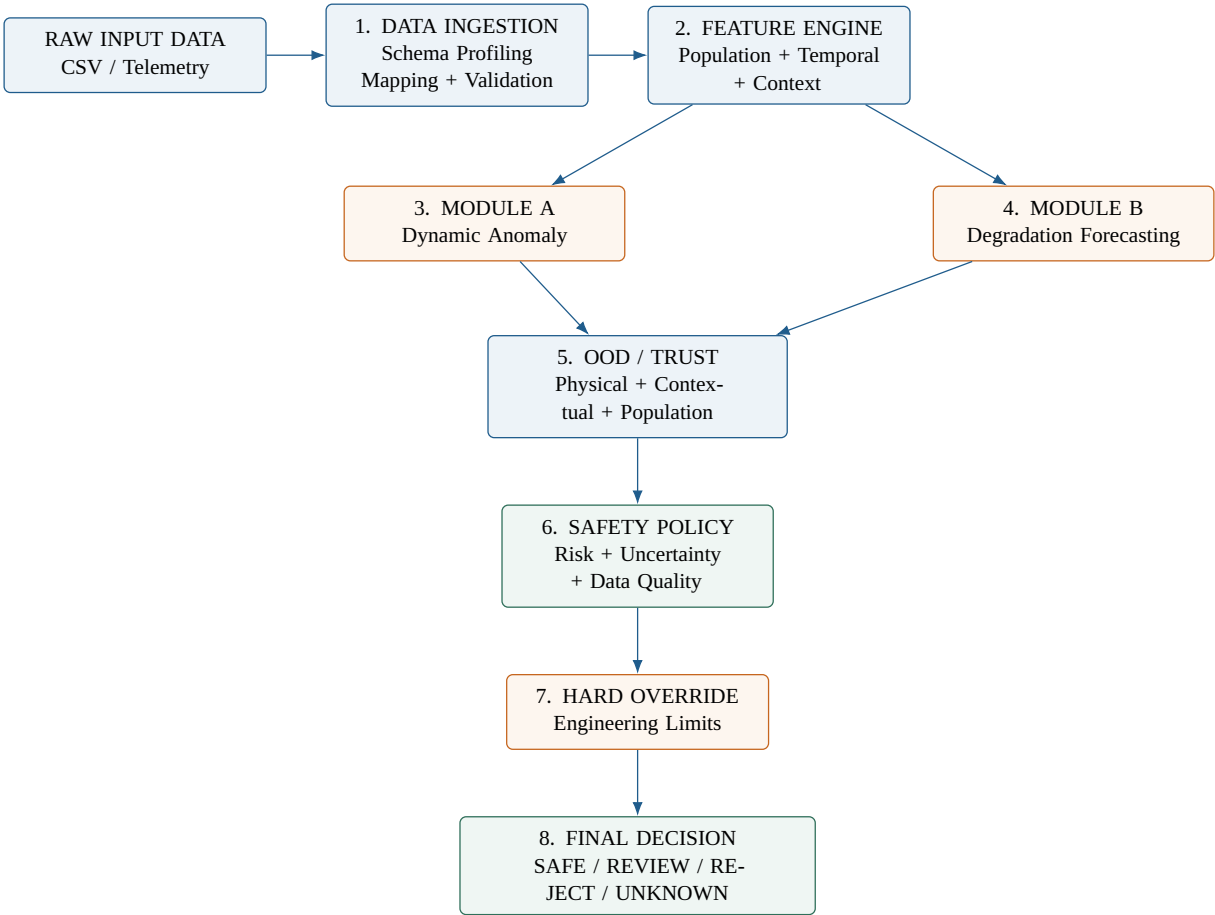
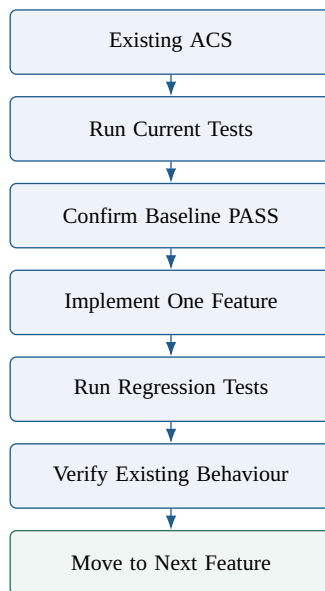


Figure 2: V4.0 implementation architecture: existing analytical foundations are extended by trust, safety and engineering-intelligence layers.



Never rewrite stable modules unnecessarily.

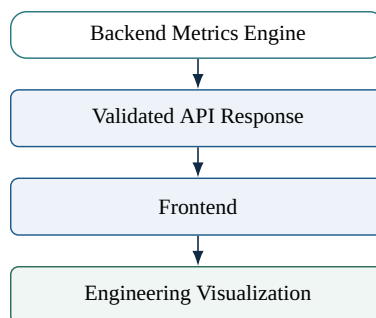
4 CENTRALIZED ENGINEERING METRICS

Create:

```
src/metrics.py
```

This module becomes the single source of truth for engineering metrics.

The frontend must NOT independently calculate engineering KPIs.

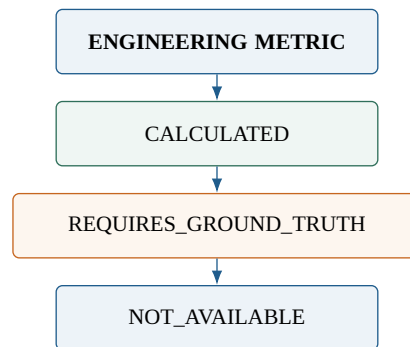


4.1 Required metrics

DPPM
 FNR
 FPR
 Precision
 Recall
 Specificity
 Critical Escape Count
 False Scrap Rate
 Review Burden
 Chamber Hours Saved
 Burn-In Reduction
 Forecast MAE
 Forecast RMSE
 Conformal Coverage
 Limit-Crossing Recall
 OOD Flag Rate
 Drift Velocity
 Lead Time

5 METRIC STATUS SYSTEM

Every metric must have an explicit status.



This prevents the UI from displaying unsupported values as real engineering measurements.

Example:

```

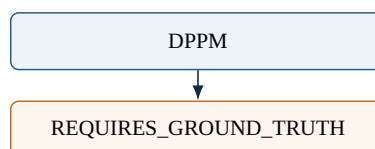
{
  "metric": "post_screening_dppm",
  "value": null,
  "status": "REQUIRES_GROUND_TRUTH"
}
  
```

6 DPPM

Defects Per Million:

$$DPPM = \frac{\text{Uncaught Defects}}{\text{Total Accepted Components}} \times 10^6$$

If validated ground truth is unavailable:



Never fabricate DPPM.

7 LATENT ESCAPE FNR

$$FNR = \frac{FN}{TP + FN}$$

Equivalent:

$$Latent\ Escape\ FNR = 1 - Latent\ Escape\ Recall$$

Safety objective:

$$FNR \rightarrow 0$$

8 CRITICAL ESCAPE COUNT

$$CriticalEscapeCount = \#(\text{missed components violating critical limits})$$

Target:

$$CriticalEscapeCount = 0$$

9 FALSE SCRAP RATE

$$FalseScrapRate = \frac{False\ Positives}{Total\ Healthy\ Components} \times 100$$

This measures unnecessary escalation/rejection of healthy components.

10 REVIEW BURDEN

$$ReviewBurden = \frac{Components\ Routed\ to\ REVIEW}{Total\ Components} \times 100$$

This measures the amount of human engineering intervention required.

11 CHAMBER HOURS SAVED

For each correctly identified component:

$$SavedHours_i = T_{conventional} - T_{ACS}$$

Total:

$$ChamberHoursSaved = \sum_{i=1}^N (T_{conventional} - T_{ACS,i})$$

For a 168 h conventional burn-in and 24 h early screening:

$$168 - 24 = 144h$$

12 BURN-IN REDUCTION

$$BurnInReduction = \frac{168 - 24}{168} \times 100$$

Therefore:

$BurnInReduction = 85.7\%$

This represents a prototype/benchmark scenario and must NOT be presented as a validated production saving.

13 FORECAST METRICS

13.1 MAE

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

13.2 RMSE

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

14 CONFORMAL PREDICTION INTERVAL

Given:

$$\hat{y}$$

and calibrated uncertainty:

$$q$$

calculate:

$$\begin{aligned} L &= \hat{y} - q \\ U &= \hat{y} + q \end{aligned}$$

Therefore:

$PredictionInterval = [L, U]$

Coverage:

$$Coverage = \frac{\#\{y_i \in [L_i, U_i]\}}{n}$$

Important:

Nominal 95%
!=
Guaranteed 95%

The system must report actual empirical coverage.

15 LIMIT-CROSSING RECALL

For upper engineering limit:

$$U_{168} > L_{max}$$

indicates potential future crossing.

For lower limit:

$$L_{168} < L_{min}$$

indicates potential future crossing.

Recall:

$$LimitCrossingRecall = \frac{TP_{cross}}{TP_{cross} + FN_{cross}}$$

16 DRIFT VELOCITY

$$DriftVelocity = \frac{\Delta x}{\Delta t}$$

For the first 24 hours:

$$Drift_{0 \rightarrow 24} = \frac{x_{24} - x_0}{24}$$

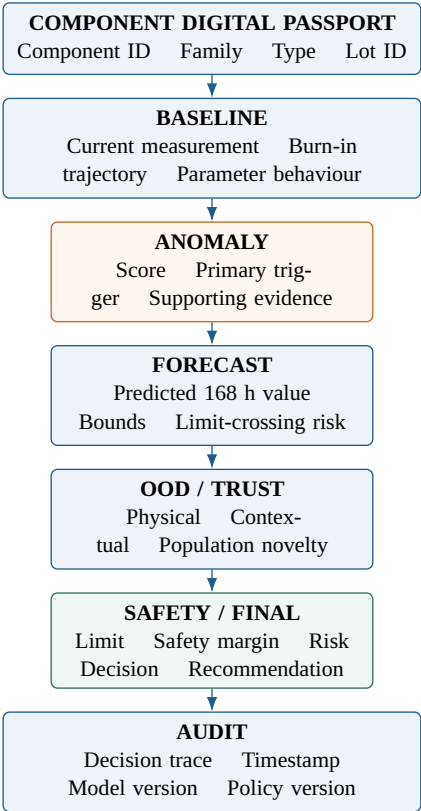
Units depend on the parameter.

Examples:

uA/h
V/h
Ohm/h
pF/h

17 COMPONENT DIGITAL PASSPORT

Create a complete component-level engineering record.



18 SAFETY MARGIN

For an upper limit:

$$SafetyMargin = L_{max} - x$$

For a lower limit:

$$SafetyMargin = x - L_{min}$$

Forecast margin:

$$ForecastMargin = L_{max} - \hat{x}_{168}$$

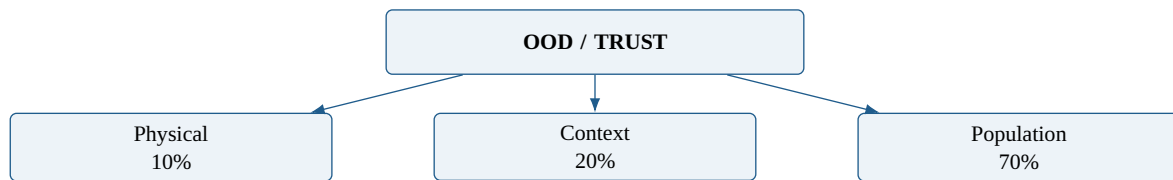
If the engineering limit is unavailable:



Never invent an engineering limit.

19 OOD V4 DECOMPOSITION

OOD must be decomposed into:



Conceptually:

$$OOD = 0.10P + 0.20C + 0.70N$$

where:

- (P) = physical novelty
- (C) = contextual novelty
- (N) = population/trajectory novelty

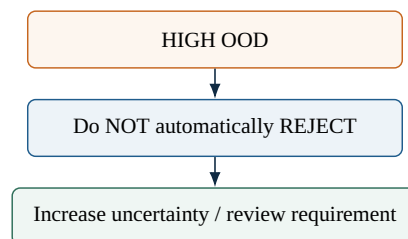
The weights are configurable engineering parameters, not universal constants.

20 OOD DECISION RULE

Critical rule:

$$OOD \neq DEFECT$$

Therefore:



Possible states:

LOW
MODERATE
SEVERE

21 ENGINEERING RECOMMENDATION ENGINE

Final decisions should include a recommended action.

Possible recommendations:

RELEASE_SAFE
CONTINUE_MONITORING
ENGINEERING_REVIEW
ADDITIONAL_SCREENING
HOLD
REJECT

Example:

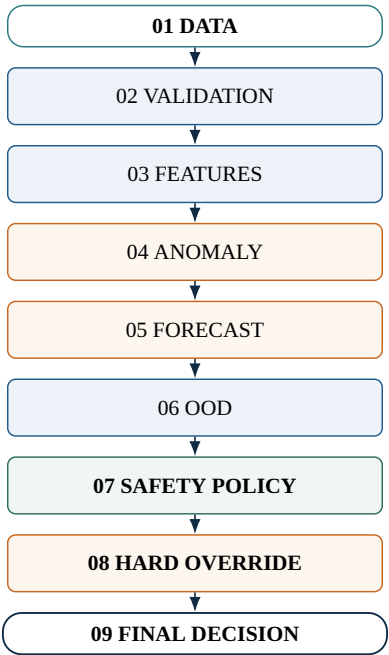
Decision:
REVIEW

Recommendation:
ENGINEERING_REVIEW

Reason:
High population anomaly + elevated forecast uncertainty

22 9-STAGE DECISION TRACE

Every component decision must expose:



Each stage must provide:

Stage
Status
Evidence
Input
Output
Reason
Timestamp

23 STAGE 01 — DATA

Verify:

- component ID,
- lot ID,
- parameter,
- measurement,
- timestamp,
- test stage.

Output:

DATA_VALID

or:

DATA_INVALID

24 STAGE 02 — VALIDATION

Check:

- schema,
- data types,
- missing values,
- units,
- measurement validity,
- physical consistency.

Output:

VALID
INVALID
PARTIAL

25 STAGE 03 — FEATURES

Generate:

Population Features
Temporal Features
Context Features
Multivariate Features

No future information may be used.

Information boundary:

$$F(t_o) \subseteq \sigma(\{x(t) : t \leq t_o\})$$

26 STAGE 04 — ANOMALY

Module A produces:

Absolute Limit Evidence
Population Evidence
Multivariate Evidence
Temporal Evidence

These are fused into anomaly evidence.

27 STAGE 05 — FORECAST

Module B produces:

Predicted 168h Value
Lower Prediction Bound
Upper Prediction Bound
Drift Estimate
Limit-Crossing Risk

28 STAGE 06 — OOD

Evaluate:

Physical Novelty
Contextual Novelty
Population Novelty

Return:

LOW
MODERATE
SEVERE

29 STAGE 07 — SAFETY POLICY

Combine:

Anomaly
Forecast Risk
Uncertainty
OOD
Data Quality
Engineering Context

into a policy decision.

30 STAGE 08 — HARD OVERRIDE

Absolute engineering limit violations must override all AI evidence.

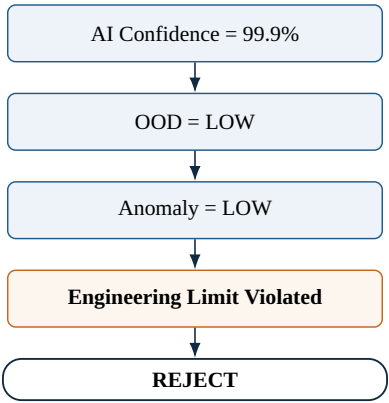
Upper limit:

$$x > L_{max} \Rightarrow REJECT$$

Lower limit:

$$x < L_{min} \Rightarrow REJECT$$

This rule is non-overridable.



31 STAGE 09 — FINAL DECISION

Allowed dispositions:

SAFE
REVIEW
REJECT
UNKNOWN

32 FINAL DECISION MATRIX

CONDITION	ACTION
Healthy + Trusted	SAFE
Moderate Risk	REVIEW
Severe Risk	REVIEW / REJECT*
Hard Limit Violation	REJECT
Severe OOD	REVIEW
Insufficient Evidence	UNKNOWN
Poor Data Quality	UNKNOWN / REVIEW

*Final action depends on validated safety policy.

33 LOT-LEVEL INTELLIGENCE

Extend analysis from individual components to manufacturing/test lots.

For each lot calculate:

Lot Size
SAFE Count
REVIEW Count
REJECT Count
UNKNOWN Count
Anomaly Rate
OOD Rate
Parameter Drift
Lot Risk
Recommended Action

Lot-level actions:

RELEASE_LOT
ELEVATED_SURVEILLANCE
QUARANTINE_LOT_HIGH_ANOMALY
ENGINEERING_REVIEW

34 LOT INTELLIGENCE LOGIC

Lot-level risk is derived from both component decisions and aggregate population behaviour, as shown in Figure 3.

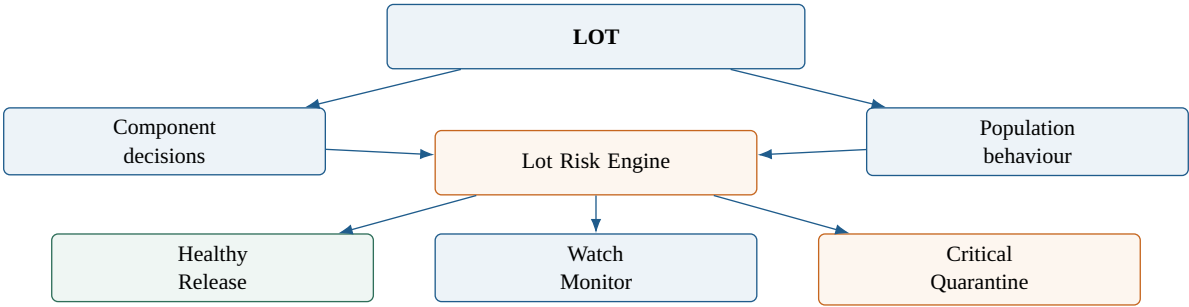


Figure 3: Lot intelligence logic: component decisions and population behaviour are fused into a lot-level risk disposition.

A single component anomaly should not automatically be interpreted as a manufacturing-lot failure.

35 FRONTEND ARCHITECTURE

The frontend must contain five primary modules:

+	-----	+
	ACS CONSOLE	
+	-----	+
	1. MISSION OVERVIEW	
	2. COMPONENT INTELLIGENCE	
	3. EVIDENCE & EXPLAINABILITY	
	4. AUDIT & BENCHMARK	
	5. NEW SCREENING	
+	-----	+

36 MISSION OVERVIEW

Dashboard KPIs:

Total Components
SAFE
REVIEW
REJECT
UNKNOWN
Escape Metrics
Review Burden
Chamber Efficiency
OOD Rate
Data Integrity

Also provide:

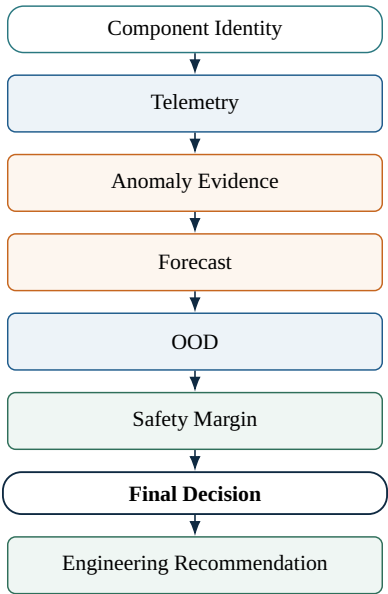
LOT EARLY WARNING

with:

- lot health,
- anomaly concentration,
- OOD behaviour,
- yield,
- quarantine recommendations.

37 COMPONENT INTELLIGENCE

The Component Intelligence screen must show:



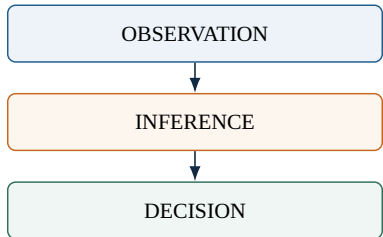
The Digital Passport is the main artifact.

38 EVIDENCE & EXPLAINABILITY

Show the 9-stage decision trace. A worked example is given in Figure 4.

39 EXPLAINABILITY STRUCTURE

Every decision must separate:



Example:

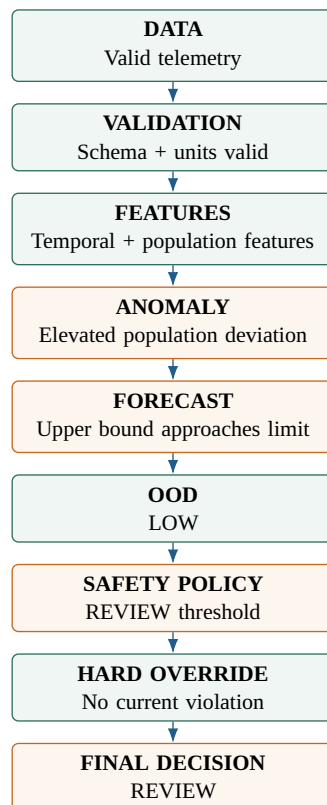


Figure 4: Worked example of a 9-stage decision trace, from raw data through to the final REVIEW disposition.

OBSERVATION:
Current leakage is 4.2x the lot median.

INFERENCE:
The component exhibits abnormal population behaviour.

DECISION:
Route to engineering review.

Avoid claiming:

"ML proved the component will fail."

Instead:

"Forecast evidence indicates elevated future risk."

40 COUNTERFACTUAL EXPLANATION

The system should answer:

What change could have weakened the current decision?

Examples:

Lower drift
+
Lower forecast uncertainty
+
Closer alignment with healthy population
=
Lower risk / possible SAFE

Counterfactuals are sensitivity analyses and must not be represented as causal proof.

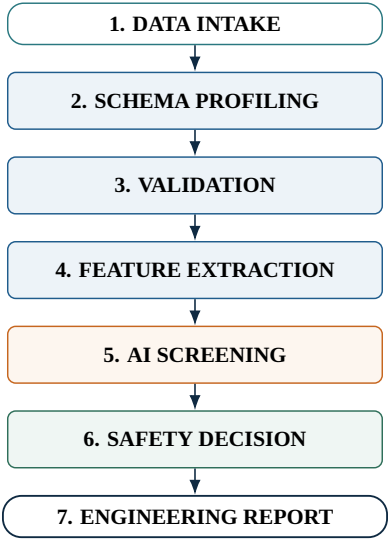
41 AUDIT & BENCHMARK

Audit page must display:

Anomaly Detection Performance
Forecast Accuracy
Limit-Crossing Recall
Latent Escape FNR
Progressive Detection
Lead Time
Conformal Coverage
Sensor-Loss Robustness
OOD Rate
Review Burden

42 NEW SCREENING WORKFLOW

The New Screening interface must provide:



43 API EXTENSIONS

Required endpoints:

```
GET /api/health
GET /api/contract

GET /api/summary

GET /api/components
GET /api/components/{id}
GET /api/components/{id}/explanation
GET /api/components/{id}/progressive
GET /api/components/{id}/passport

GET /api/lots
GET /api/lots/{lot_id}

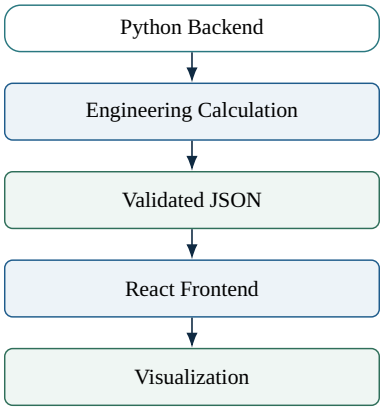
GET /api/metrics/engineering

GET /api/audit/benchmark

POST /api/screen
POST /api/intake/profile
```

44 API RESPONSE PRINCIPLE

Backend is the source of truth.



The frontend must not independently recalculate:

```
DPPM
FNR
Safety Margin
OOD
Forecast Metrics
Decision Logic
```

45 EXAMPLE COMPONENT API RESPONSE

```
{
  "component_id": "C-1024",
  "lot_id": "LOT-07",
  "decision": "REVIEW",

  "anomaly": {
    "score": 0.81,
    "status": "ELEVATED",
    "primary_trigger": "POPULATION_DEVIATION"
  },

  "forecast": {
    "prediction_168h": 47.2,
    "lower": 43.8,
    "upper": 51.1,
    "limit_crossing": true
  },

  "ood": {
    "physical": 0.04,
    "context": 0.12,
    "population": 0.38,
    "status": "MODERATE"
  },

  "safety": {
    "current_margin": 5.1,
    "forecast_margin": 2.8,
    "hard_limit_violation": false
  },

  "recommendation": "ENGINEERING_REVIEW"
}
```

46 AUDIT RECORD

Each screening decision should preserve:

- Component ID
- Lot ID
- Input Dataset
- Schema Version
- Feature Version
- Model Version
- OOD Version
- Policy Version
- Engineering Limits Version
- Decision
- Recommendation
- Decision Trace
- Timestamp

This creates reproducibility.

47 MODEL VERSIONING

Every prediction must be associated with:

Model Name
Model Version
Training Dataset Version
Feature Version
Calibration Version
OOD Reference Version
Safety Policy Version

Example:

Model:
ACS-Forecast

Version:
4.0.0

Feature Engine:
4.0

OOD:
V4

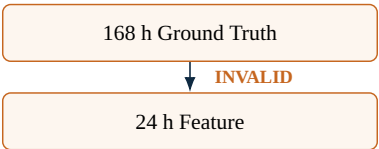
Policy:
4.0

48 INFORMATION LEAKAGE PREVENTION

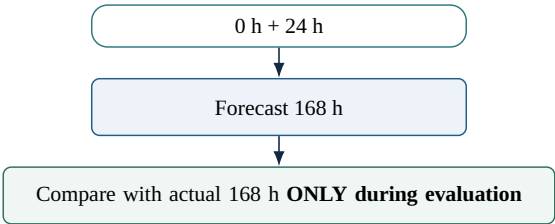
At origin (t_o):

$$F(t_o) \subseteq \sigma(\{x(t) : t \leq t_o\})$$

Never allow:



Correct:



49 PROGRESSIVE SCREENING

Evaluate at:

12h
24h
48h
72h
96h
120h
144h
168h

Lead time:

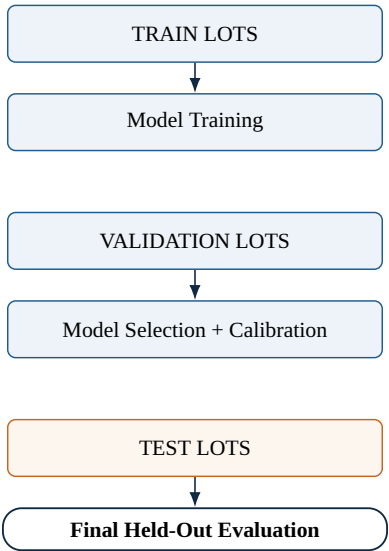
$$LeadTime = T_c - T_a$$

where:

- (T_c) = conventional detection time
- (T_a) = ACS detection time

50 TRAIN / VALIDATION / TEST SPLIT

Use lot-based separation.



Constraint:

$$TrainLots \cap TestLots = \emptyset$$

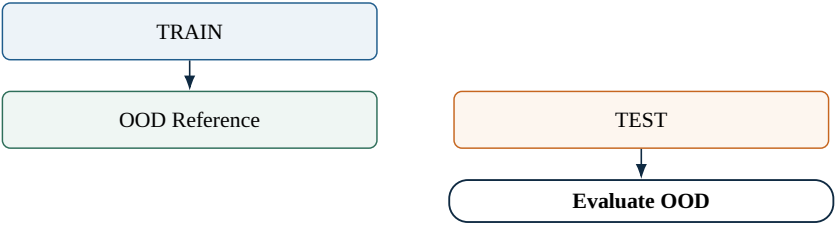
and:

$$ValidationLots \cap TestLots = \emptyset$$

51 OOD REFERENCE DATA

OOD reference distributions must be built from training/reference data.

Never construct OOD reference statistics using the final test set.



52 SYNTHETIC BENCHMARK

Support controlled failure scenarios:

Healthy Stable
Hard Early Failure
Latent Abrupt Failure
Accelerating Degradation
Change-Point Failure
Normal Aging

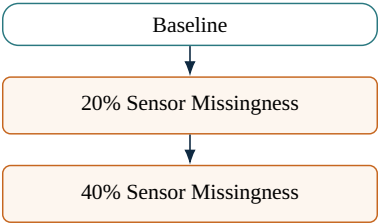
Telemetry can include:

Tester Noise
Quantization
Repeated Measurements
Missingness
Censoring

Ground-truth labels must remain evaluation-only.

53 ROBUSTNESS TESTING

Evaluate:



Metrics:

$$\Delta F1 = F1_{baseline} - F1_{missing}$$
$$\Delta Recall = Recall_{baseline} - Recall_{missing}$$

The system should explicitly show degradation rather than hiding it.

54 AUTOMATED TESTING

Target:

18/18 TESTS PASS

Test categories:

```
SAFETY POLICY
+-- Hard-limit override
+-- OOD isolation
+-- SAFE behaviour
+-- Uncertainty routing
+-- Trace consistency
+-- Safety margin
```

```
PIPELINE
+-- DPPM
+-- FNR
+-- Chamber hours
+-- Review burden
+-- Schema errors
+-- Empty input
```

```
API
+-- Health
+-- Summary
+-- Components
+-- Passport
+-- Lots
+-- Metrics
+-- Contract
```

55 SAFETY REGRESSION TEST

Mandatory test:

```
Input:
AI predicts SAFE

Condition:
Authoritative engineering limit violated

Expected:
REJECT
```

Pseudo-test:

```
def test_hard_limit_cannot_be_overridden():
    evidence = {
        "ai_decision": "SAFE",
        "anomaly_score": 0.01,
        "ood": "LOW"
    }

    result = apply_safety_policy(
        value=105,
        upper_limit=100,
        evidence=evidence
    )

    assert result.decision == "REJECT"
```

56 OOD REGRESSION TEST

Mandatory test:

Input:
Severe OOD

Condition:
No defect evidence
No hard limit violation

Expected:
NOT automatic REJECT

Expected:

REVIEW

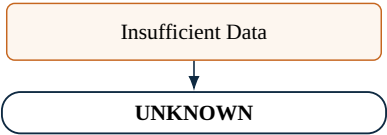
or:

UNKNOWN

depending on evidence completeness.

57 DATA QUALITY SAFETY

If the system cannot establish trustworthy evidence:



The system must not infer:

UNKNOWN -> SAFE

simply because no anomaly was detected.

58 UI DATA INTEGRITY

The frontend must clearly distinguish:

CALCULATED

from:

REQUIRES_GROUND_TRUTH

and:

NOT_AVAILABLE

Example:

Post-Screening DPPM
REQUIRES GROUND TRUTH

rather than:

DPPM = 0

when the dataset cannot establish that result.

59 PERFORMANCE CLAIM POLICY

The final UI/report must separate:

VALIDATED BENCHMARK RESULT

from:

PROTOTYPE SCENARIO

from:

REQUIRES OPERATIONAL DATA

Example:

85.7% Burn-In Reduction
STATUS: PROTOTYPE SCENARIO

DPPM
STATUS: REQUIRES GROUND TRUTH

Forecast MAE
STATUS: BENCHMARK VALIDATED

60 ENGINEERING DECISION PHILOSOPHY

The system must preserve:

$$Observation \rightarrow Inference \rightarrow Decision$$

Never:

$$ModelScore \rightarrow AutomaticTruth$$

61 COMPLETE ACS DECISION FLOW

Figure 5 summarizes the complete, end-to-end ACS V4.0 decision flow.

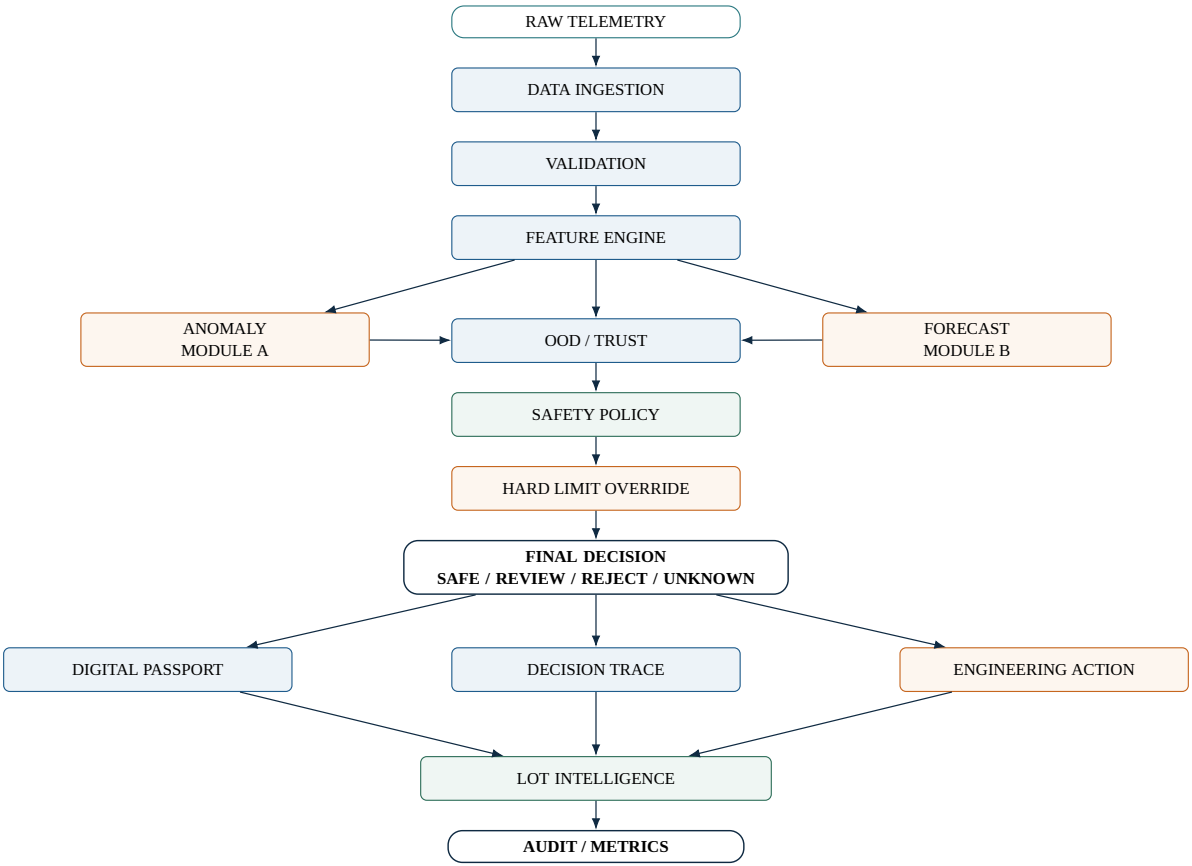
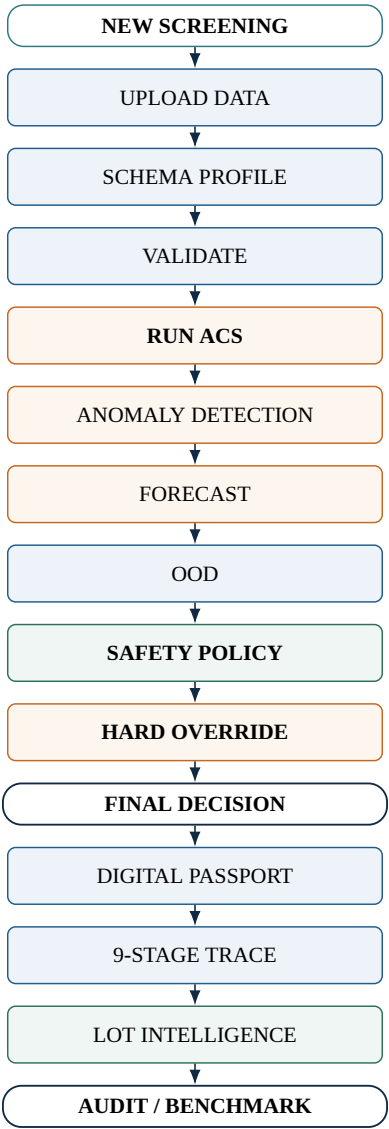


Figure 5: Complete ACS V4.0 decision flow, from raw telemetry to lot intelligence and audit.

62 SIH DEMONSTRATION FLOW

The live demonstration should follow:



63 DEMONSTRATION SCENARIO

63.1 Scenario A — Healthy Component

```
Normal telemetry
+
Low anomaly
+
Low OOD
+
Safe forecast
+
Healthy safety margin
=
SAFE
```

63.2 Scenario B — Latent Degradation

Normal current value
+
Abnormal drift
+
High population deviation
+
Elevated forecast risk
=
REVIEW

63.3 Scenario C — OOD Component

Unfamiliar operating context
+
No hard-limit violation
+
No strong defect evidence
=
REVIEW / UNKNOWN

63.4 Scenario D — Hard Limit Violation

AI Evidence:
SAFE

BUT

Engineering Limit:
VIOLATED

Therefore:

REJECT

This scenario should be included in the SIH demo because it visibly proves the safety architecture.

64 FINAL ENGINEERING KPI BOARD

ACS ENGINEERING KPIs	
Total Components	
SAFE / REVIEW / REJECT / UNKNOWN	
Latent Escape FNR	
Critical Escape Count	
DPPM	
False Scrap Rate	
Review Burden	
Forecast MAE	
Forecast RMSE	
Conformal Coverage	
Limit-Crossing Recall	
OOD Flag Rate	
Average Lead Time	
Chamber Hours Saved	
Burn-In Reduction	

65 FINAL IMPLEMENTATION CHECKLIST

65.1 Backend

- ☐ `src/metrics.py`
- ☐ Digital Passport service
- ☐ Safety Margin calculation
- ☐ 9-stage Decision Trace
- ☐ OOD V4 decomposition
- ☐ Engineering Recommendation
- ☐ Lot intelligence
- ☐ Engineering KPI API
- ☐ Passport API
- ☐ Lot API
- ☐ Screening API
- ☐ Schema profiling API

65.2 Frontend

- ☐ Mission Overview
- ☐ Component Intelligence
- ☐ Digital Passport
- ☐ Evidence & Explainability
- ☐ 9-stage Trace
- ☐ Audit & Benchmark
- ☐ New Screening
- ☐ Lot Intelligence
- ☐ Engineering KPI cards
- ☐ Metric status indicators
- ☐ Safety Margin visualization
- ☐ OOD decomposition

65.3 Testing

- ☐ Safety policy tests
- ☐ Hard override test
- ☐ OOD isolation test
- ☐ Metrics tests
- ☐ API tests
- ☐ Schema tests
- ☐ Empty-input tests
- ☐ Passport tests
- ☐ Lot tests
- ☐ Regression suite

66 FINAL ACCEPTANCE CRITERIA

The implementation is considered complete when:

- ✓ Existing ACS functionality remains operational
- ✓ 18/18 automated tests pass
- ✓ Hard engineering limits cannot be overridden
- ✓ OOD does not automatically imply defect
- ✓ Every component has an explainable decision
- ✓ Every decision exposes the 9-stage trace
- ✓ Digital Passport is available
- ✓ Safety Margin is calculated when limits exist
- ✓ Missing ground truth is explicitly labelled
- ✓ DPPM is never fabricated
- ✓ Forecast uncertainty is exposed
- ✓ Lot-level intelligence is available
- ✓ New Screening workflow works end-to-end
- ✓ Backend is the source of engineering truth
- ✓ Frontend consumes validated API results
- ✓ Audit information is reproducible
- ✓ Prototype claims are clearly separated from validated/operational claims

67 FINAL SYSTEM DEFINITION

The completed ACS system can be summarized as:

$$ACS = Data + Features + Anomaly + Forecast + OOD + Safety + Explainability + Audit$$

The operational intelligence loop is:

$$DETECT \rightarrow PREDICT \rightarrow TRUST \rightarrow EXPLAIN \rightarrow DECIDE$$

The safety principle is:

$$AI\ Evidence \neq Engineering\ Authority$$

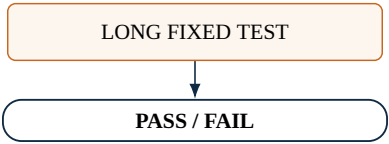
And the non-negotiable safety invariant is:

$$Authoritative\ Limit\ Violation \Rightarrow REJECT$$

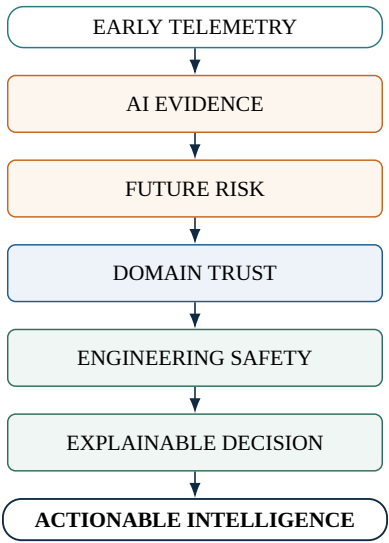
68 FINAL PROJECT POSITIONING

Adaptive Component Screening (ACS) is an AI-driven aerospace electronics screening intelligence platform that detects abnormal component behaviour, forecasts future degradation, evaluates model/domain trust, quantifies uncertainty, enforces deterministic engineering safety limits, explains every decision and provides component- and lot-level engineering intelligence.

The final system transforms conventional burn-in screening from:



into:



Therefore:

Adaptive Component Screening = Early Detection + Predictive Reliability + Safety Intelligence
ACS V4.0 — Engineering Intelligence Upgrade